

Pyrin (MEFV) versus NLRP3 programs in human carotid plaque: a hypothesis-generating map

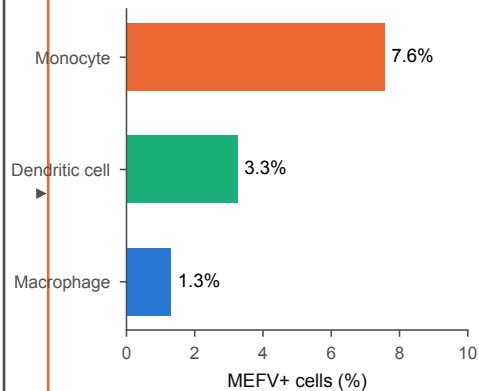
Data and controls

GSE159677 scRNA-seq
3 patients, core + adjacent
44,454 singlets

GSE120521 bulk RNA-seq
4 stable/unstable pairs

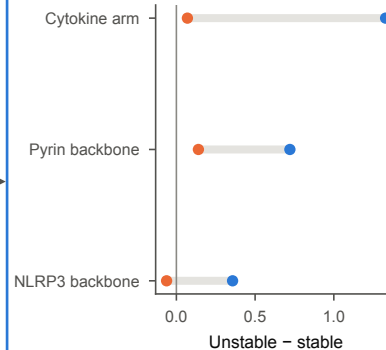
Doublets, ambient RNA,
sequencing depth controlled;
sample labels verified

Where is MEFV?



MEFV is sparse and monocyte-enriched;
rare in TREM2+ lipid-associated macrophages

What drives bulk signals?



Blue: unadjusted; orange: adjusted for myeloid
content. Bulk signals track myeloid abundance

Two-signal model

Signal 1 (priming)
→ IL-1 β magnitude and
IL-1 β :IL-18 ratio

Signal 2 / pyrin gate
→ timing of IL-1 β , IL-18
and lysis

Dimensionless, uncalibrated;
generates testable predictions

Transcript scores do not measure inflammasome activation; findings prioritise cell-type-resolved, protein-level and genotype-aware validation.